



PREDICTIVE ANALYTICS DASHBOARD FOR PEDIATRIC ASTHMA USING MACHINE LEARNING



Final Year Project | Bachelor of Computer Science (Hons.)

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Introduction

Predictive Analytics Dashboard for Pediatric Asthma is a web-based application that is used to aid in the early detection of asthma in children by machine learning methodology. It involves the application of **clinical and demographic data** to **predict** asthma by using **supervised machine learning models**.

The system **combines predictive models with interactive dashboard** interface to give **visual insights**, which help users interpret prediction outcomes.

Problem Statement

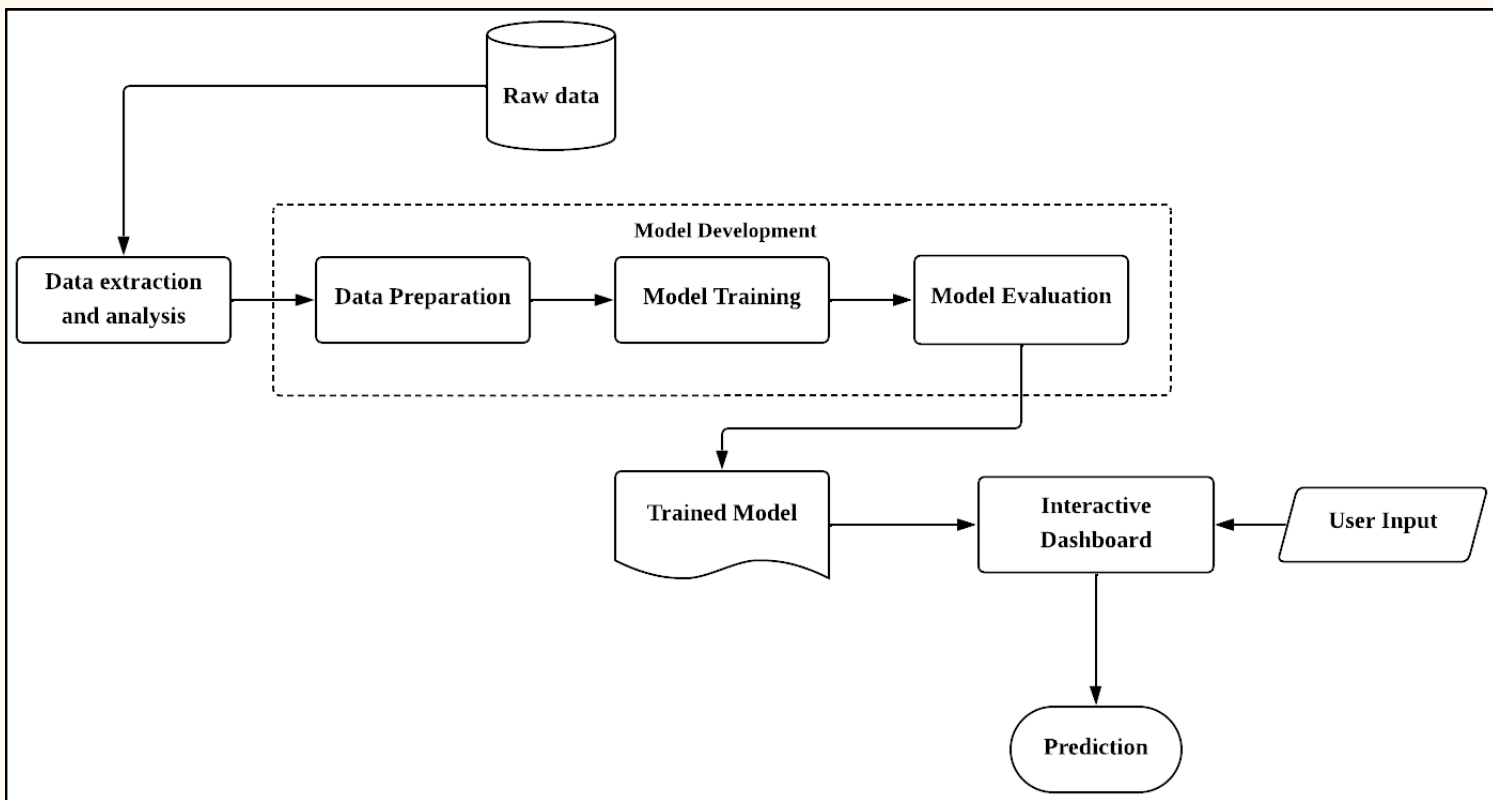
- 1.Challenges in predicting asthma in children at an **early stage**.
- 2.Unstable ML prediction with **unbalanced data**.
- 3.Absence of available predictive tools in **visual form**.

Objective

- 1.To **determine** the appropriate **machine learning model** in the prediction of pediatric asthma.
- 2.To **develop a predictive analytics dashboard** in the web-based format including visual insights.
- 3.To **test the functionality** and the performance of dashboards regarding the accuracy of the models as well as their usability and perspective of end-users.

Methodology

System Architecture



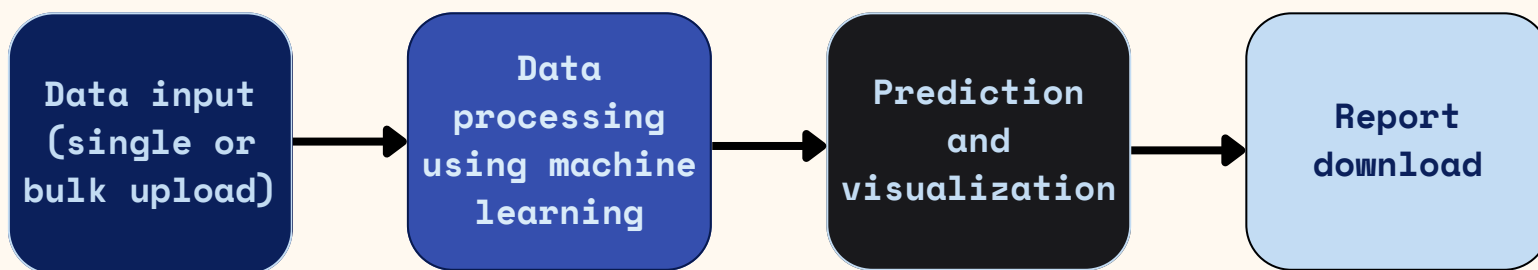
Data Collection: A public dataset containing **clinical and demographic attributes**.

Software Requirements:

- Python Libraries for ML (Scikit-learn, XGBoost libraries, Pandas, Numpy)
- Python Framework for dashboard (Streamlit)

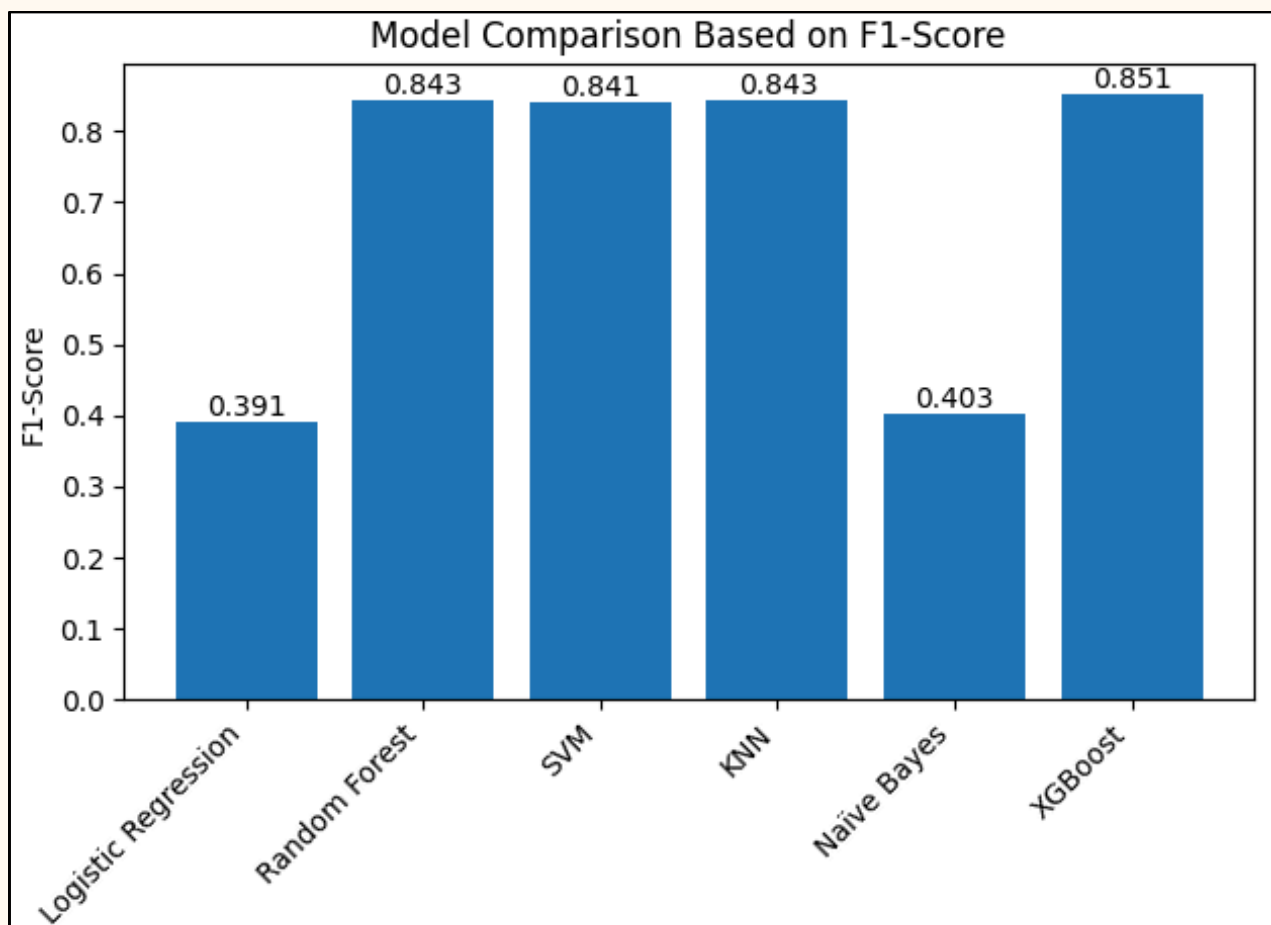
Programming Language: Python

Dashboard System Flow

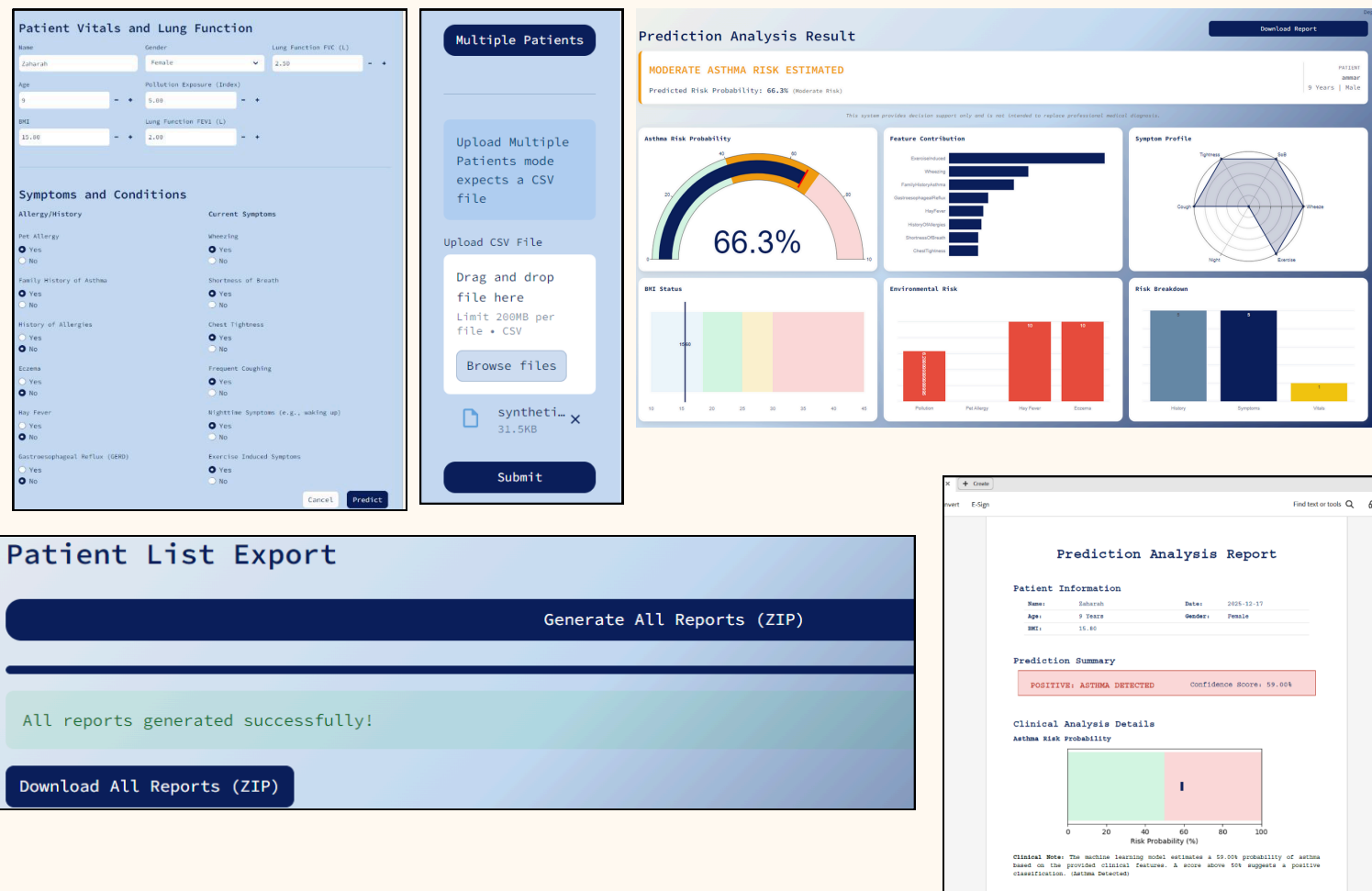


Result

XGBoost achieved the highest **F1-score (0.851)** compared to other evaluated models and was selected for dashboard integration.



Web-based Dashboard System (User Interface)



Conclusion

This project successfully **developed** a predictive analytics dashboard for pediatric asthma using supervised machine learning techniques. Various models were compared and **XGBoost had the best F1-score** making it the most appropriate model to use in prediction.

Model integration into a **web-based dashboard** with visualization features **allows users to interpret the results** of the prediction in a better manner and **aid the possibility of early asthma screening**.

Future Work

- 1.**Validate** the proposed system using **real clinical and hospital data**
- 2.**Larger** and more **diverse dataset**
- 3.**Real-time** system deployment
- 4.**Dashboard** usability **enhancement** based on user-feedback